

Principles of Anatomy and Physiology

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Sixteenth Edition

Chapter 6

The Skeletal System: Bone Tissue

Introduction

- The purpose of the chapter is to:
 - Describe the functions of the skeletal system
 - Describe the structure of long bones
 - Compare and contrast endochondral and intramembranous ossification
 - Identify histology of compact and spongy bone tissue
 - Outline the roles of hormones in bone tissue and calcium homeostasis
 - Identify common types of fractures

Functions of Bone and the Skeletal System

Bone

- Bone is an organ made up of several different tissues (bone, cartilage, dense connective tissue, adipose, and nervous tissue) working together
- The entire framework of bones and their cartilages constitute the skeletal system

Functions of the Skeletal System

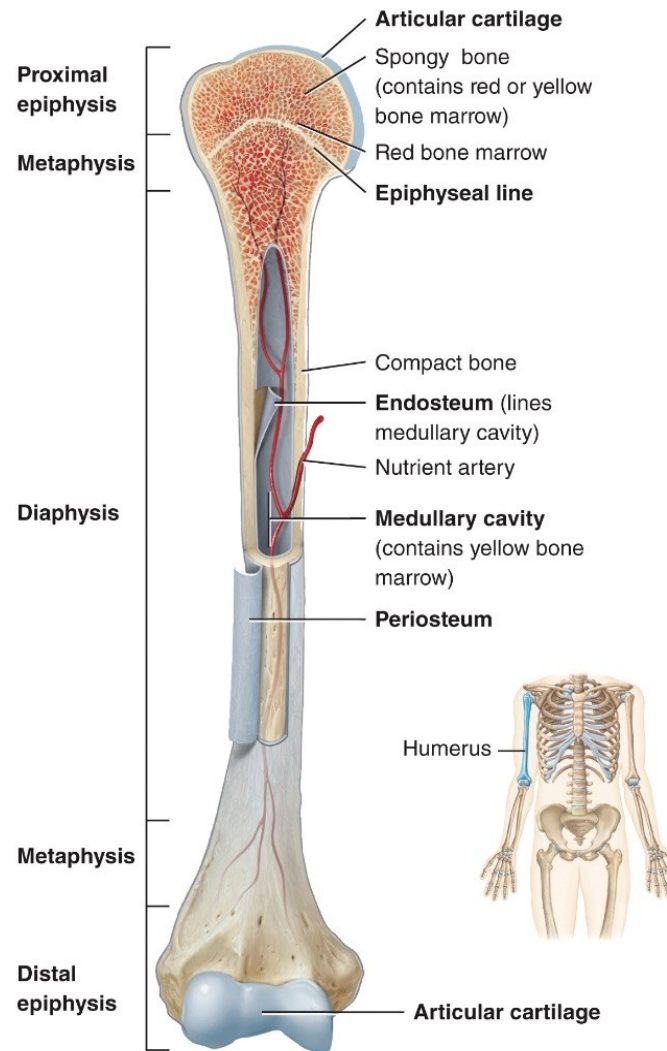
- Provides support
- Protects the internal organs (brain, heart, etc.)
- Assists body movements (in conjunction with muscles)
- Mineral homeostasis - stores and releases calcium and phosphorus
- Participates in blood cell production (hemopoiesis or hematopoeisis)
- Stores triglycerides in adipose cells of yellow marrow

Structure of Bone

Structure of Bone

- A **long bone** consists of several parts:
 - Diaphysis (bone shaft)
 - 2 epiphyses (both ends of the bone at the joints)
 - 2 metaphyses (region between diaphysis and epiphysis)
 - Articular cartilage covering both epiphyses
 - Periosteum (connective tissue surrounding the diaphysis)
 - Medullary cavity (hollow space within diaphysis)
 - Endosteum (thin membrane lining the medullary cavity)

Structure of Bone



(a) Partially sectioned humerus

Histology of Bone

Histology of Bone

- Bone contains an abundant extracellular matrix that surrounds widely separated cells
 - The extracellular matrix is about 15% water, 30% collagen, and 55% crystalized mineral salts
- Mineral salts calcium phosphate and calcium hydroxide combine to form crystals called hydroxyapatite

Histology of Bone

- Bone contains 4 types of cells:
 - **Osteoprogenitor cells** (bone stem cells able to differentiate into the other types of cells)
 - **Osteoblasts** (bone-building cells that secrete matrix; initiate calcification)
 - **Osteocytes** (mature bone cells)
 - **Osteoclasts** (remodel bones and cause them to release calcium; bone resorption)

Histology of Bone

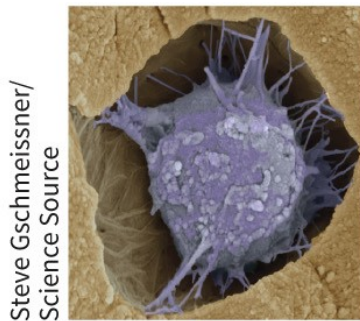


Osteoprogenitor cell
(develops into an osteoblast)

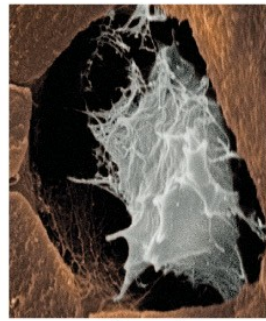
Osteoblast
(functions in bone deposition, the buildup of bone extracellular matrix)

Osteocyte
(maintains bone tissue)

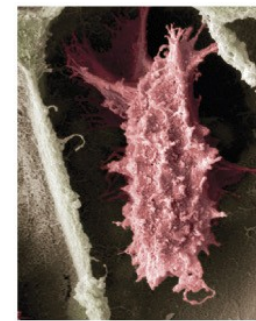
Osteoclast
(functions in bone resorption, the breakdown of bone extracellular matrix)



SEM 8000x



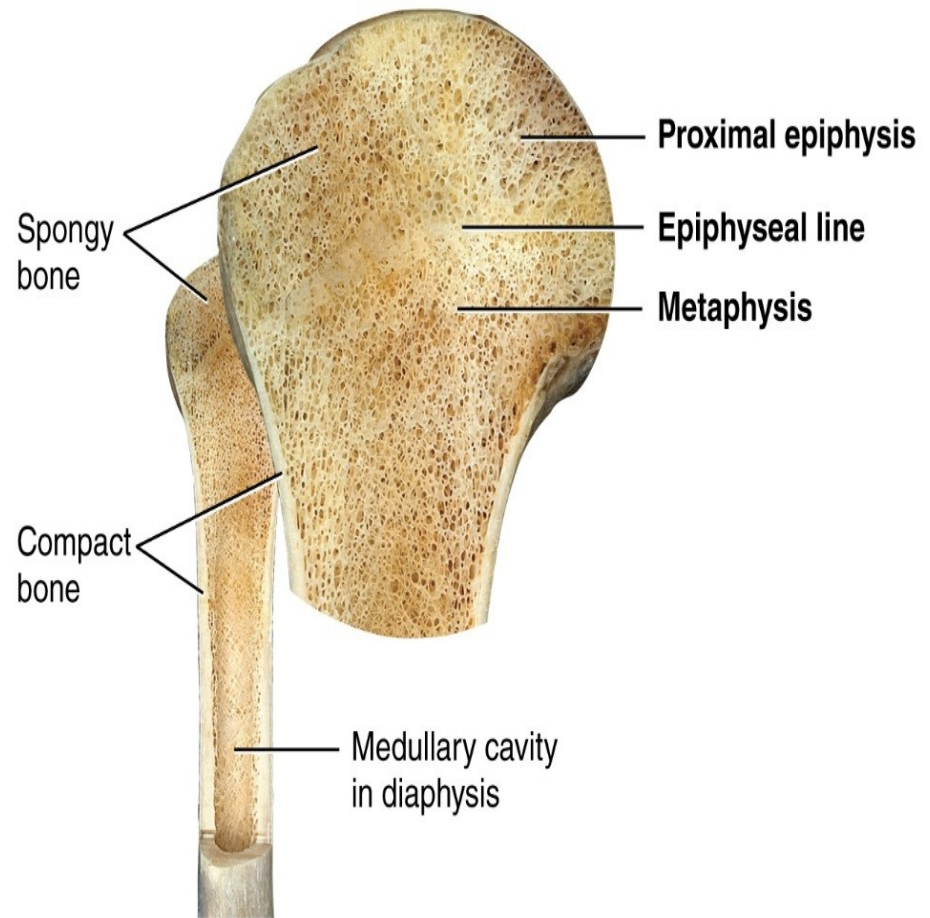
SEM 4000x



SEM 2700x

Histology of Bone

- **Compact bone** is good at providing protection and support; strongest
- **Spongy bone** is lightweight and provides tissue support; also called trabecular or cancellous bone



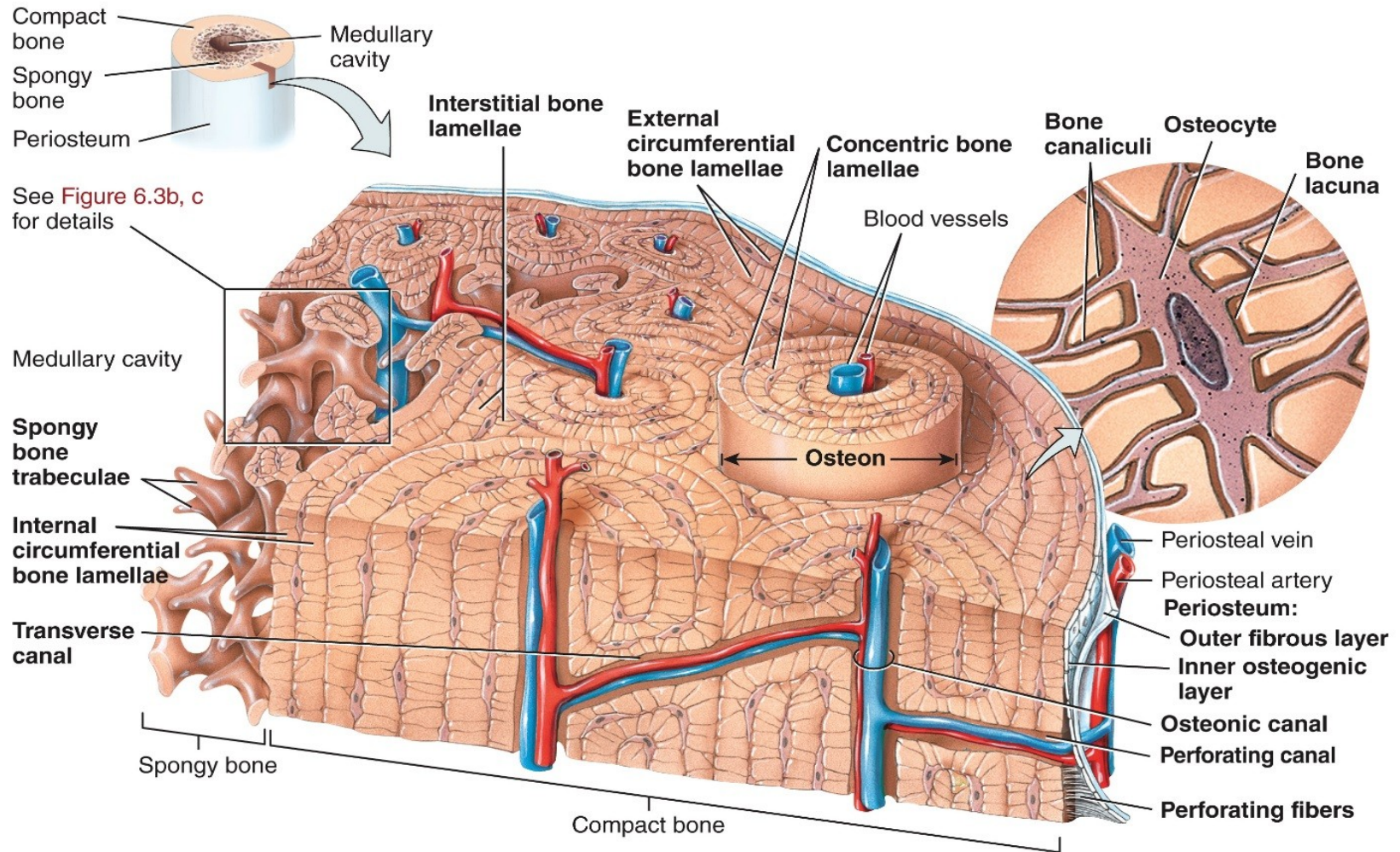
(b) Partially sectioned humerus

Histology of Bone

Compact Bone

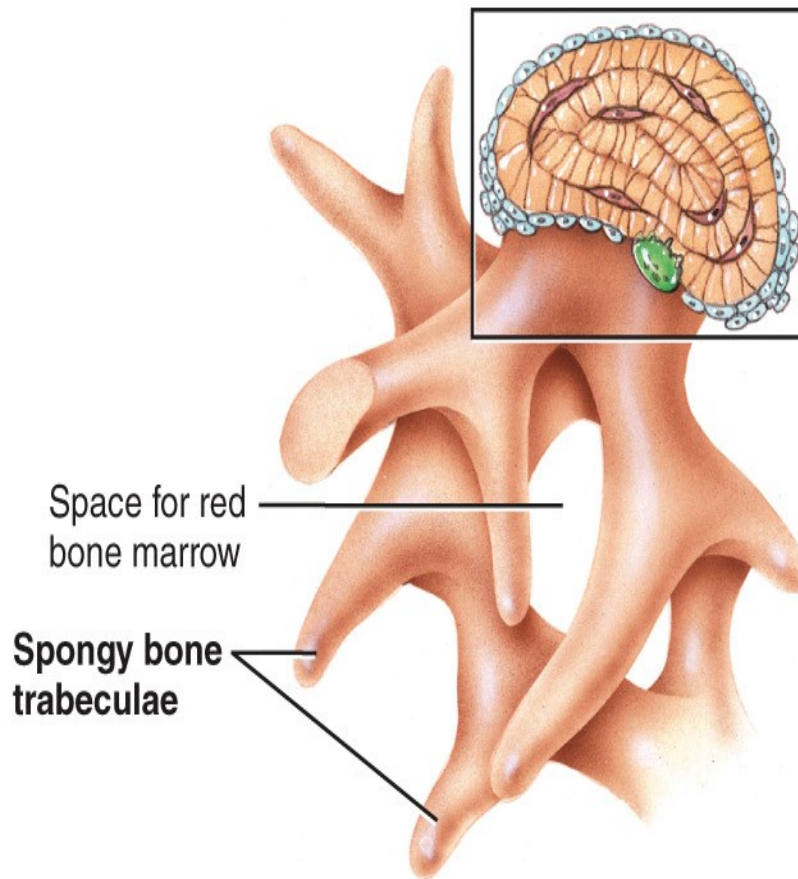
Spongy (Cancellous) Bone

Histology of Bone

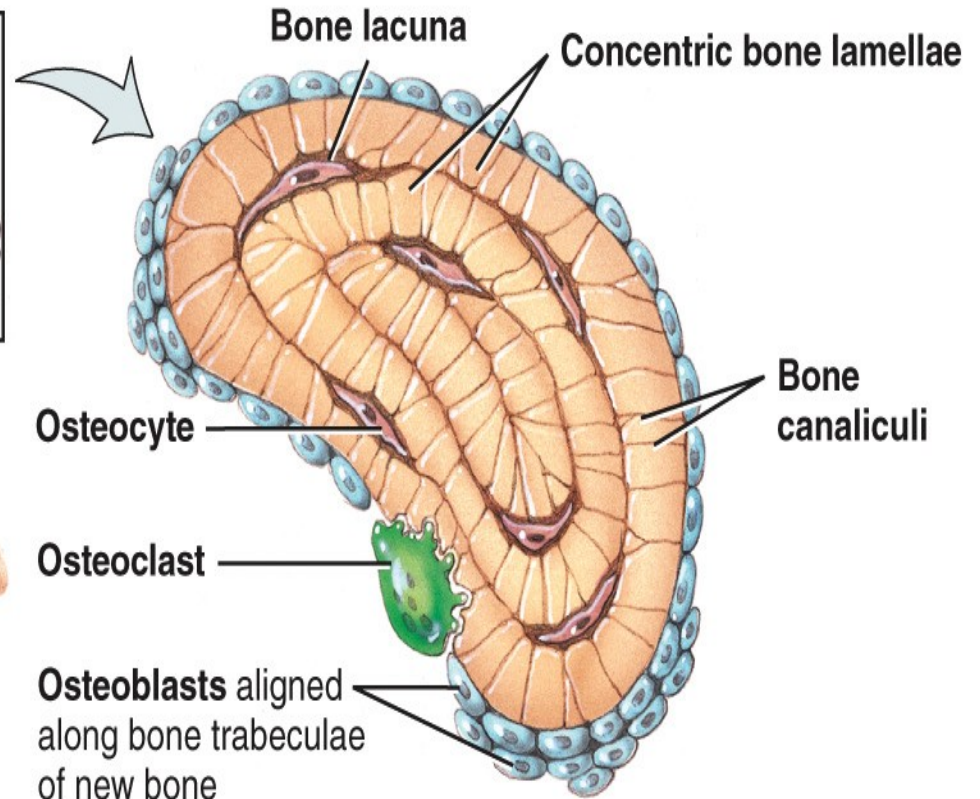


(a) Osteons in compact bone and bone trabeculae in spongy bone

Histology of Bone



(b) Enlarged aspect of spongy bone trabeculae

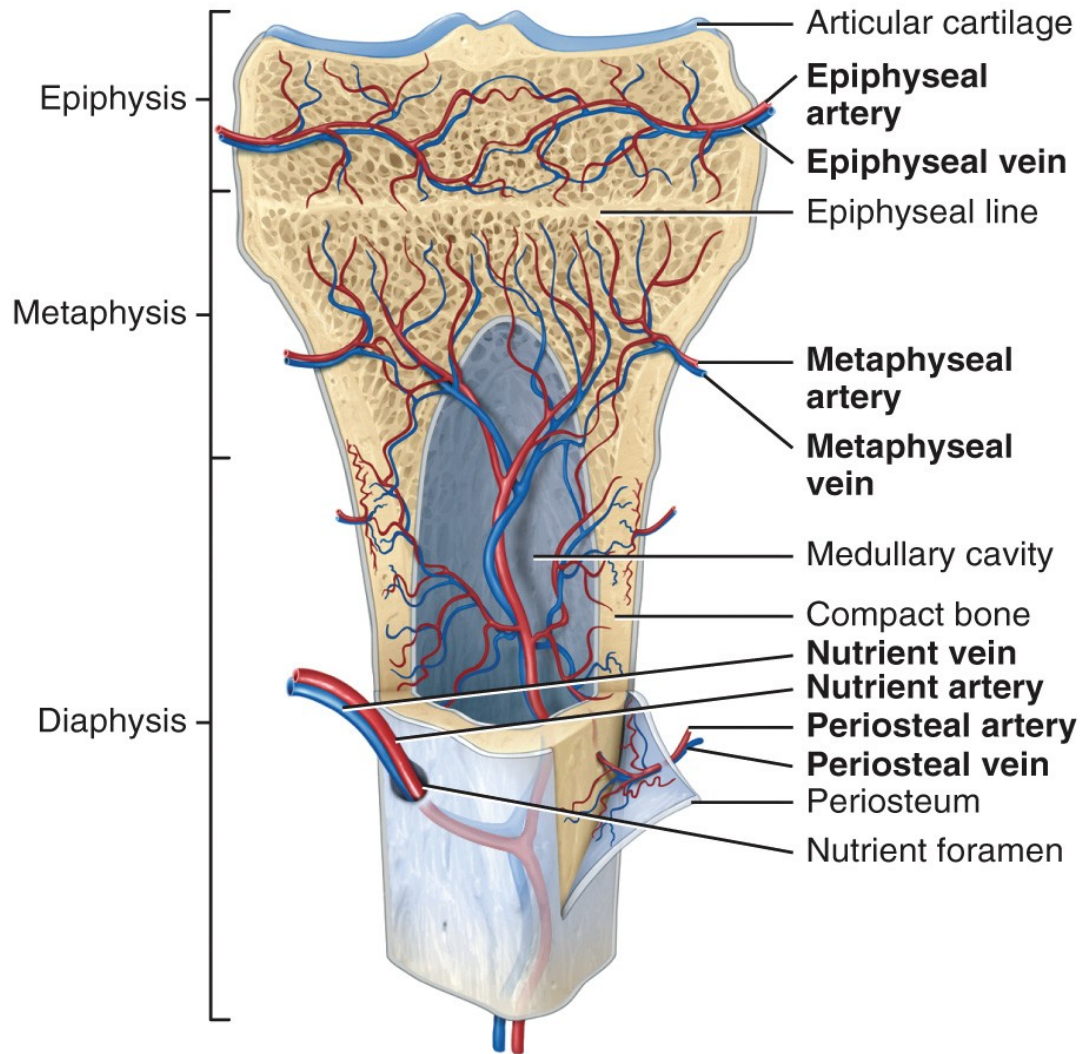


(c) Details of a section of a spongy bone trabecula

Blood and Nerve Supply of Bone

- Periosteal arteries (accompanied by nerves) enter the diaphysis through Volkmann's canals. They are accompanied by periosteal veins
- A nutrient artery enters the center of the diaphysis through a nutrient foramen. Nutrient veins exit via the same canal
- The metaphyses and epiphyses also have their own arteries and veins

Blood and Nerve Supply of Bone



Partially sectioned tibia (shin bone)

Bone Formation

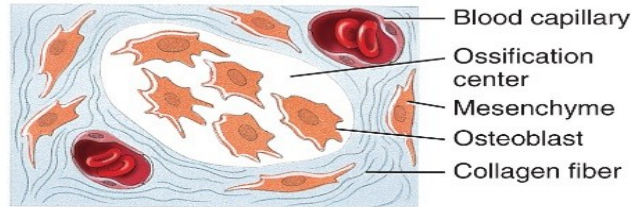
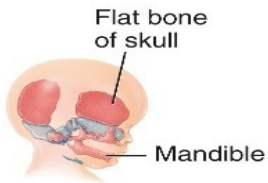
Bone Formation

- Ossification (osteogenesis) is the process of bone formation
- Bones form in 4 situations:
 - During embryological and fetal development
 - When bones grow before adulthood
 - When bones remodel
 - When fractures heal

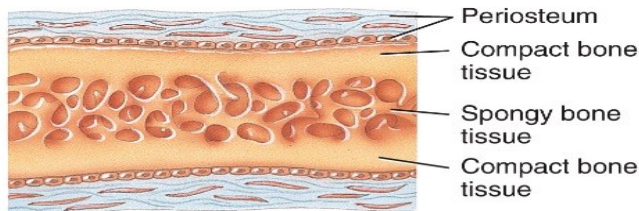
Bone Formation

- Types of ossification:
 - Intramembranous
 - Occurs in flat bones when a connective tissue membrane is replaced by bone
 - Endochondral
 - Replaces cartilage with bone in the developing embryo and fetus

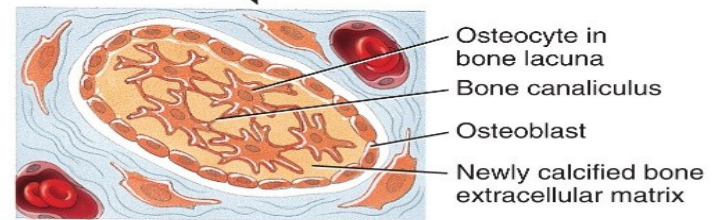
Bone Formation - Intramembranous



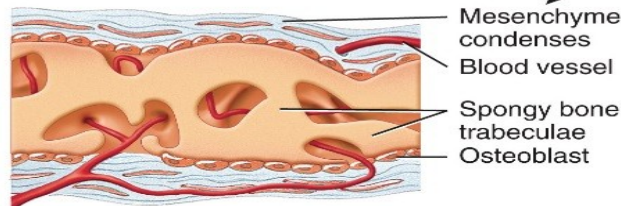
- 1 Development of ossification center:** osteoblasts secrete organic extracellular matrix.



- 4 Development of the periosteum:** mesenchyme at the periphery of the bone develops into the periosteum.

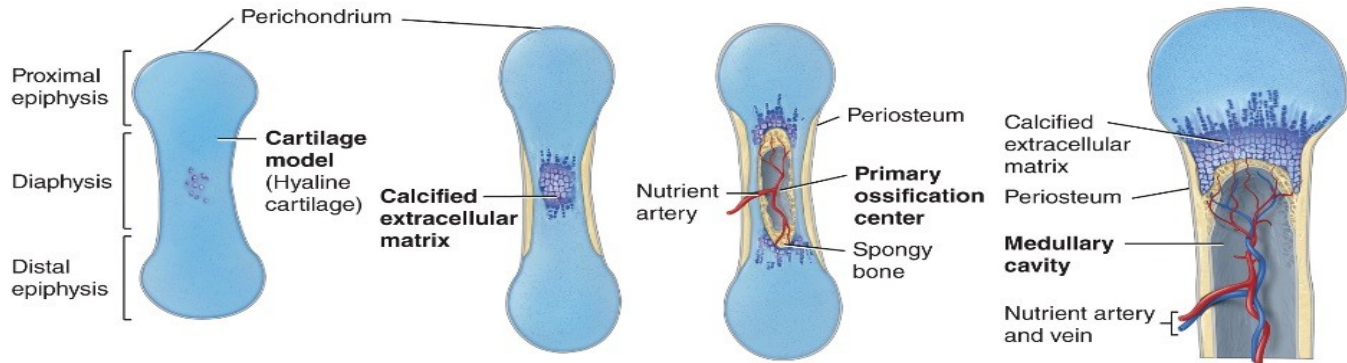


- 2 Calcification:** calcium and other mineral salts are deposited and extracellular matrix calcifies (hardens).



- 3 Formation of spongy bone trabeculae:** extracellular matrix develops into spongy bone trabeculae that fuse to form spongy bone.

Bone Formation – Endochondral

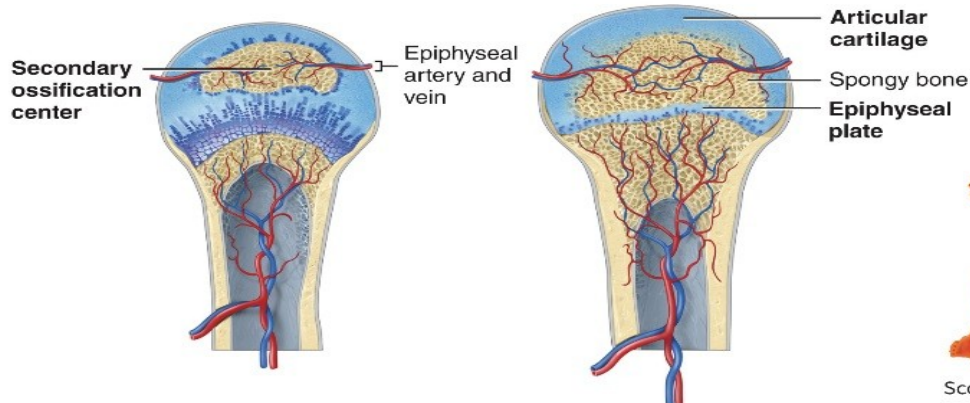


1 Development of cartilage model: Mesenchymal cells develop into chondroblasts, which form the cartilage model.

2 Growth of cartilage model: Growth occurs by cell division of chondrocytes.

3 Development of primary ossification center: In this region of the diaphysis, bone tissue has replaced most of the cartilage.

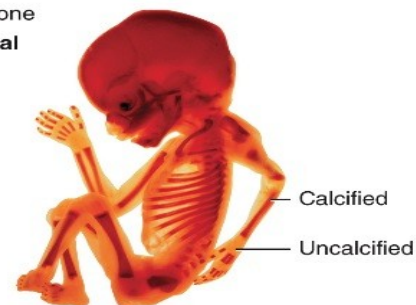
4 Development of the medullary (marrow) cavity: Bone breakdown by osteoclasts forms the medullary cavity.



5 Development of secondary ossification centers: These occur in the epiphyses of the bone.

6 Formation of articular cartilage and epiphyseal plate: Both structures consist of hyaline cartilage.

(a) Sequence of events

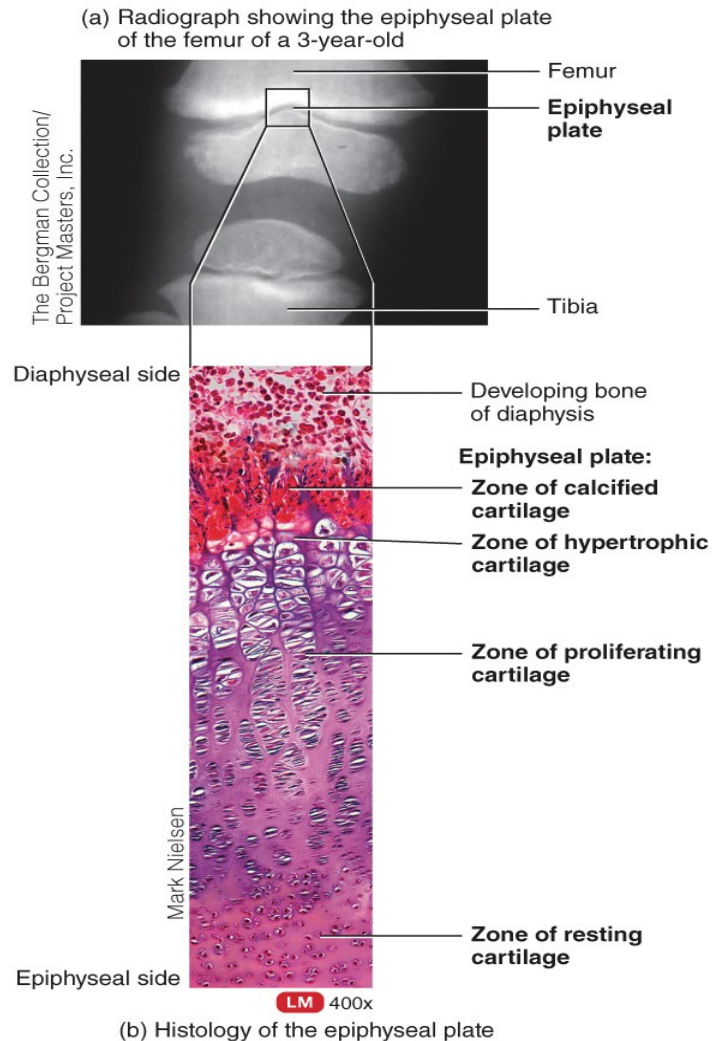
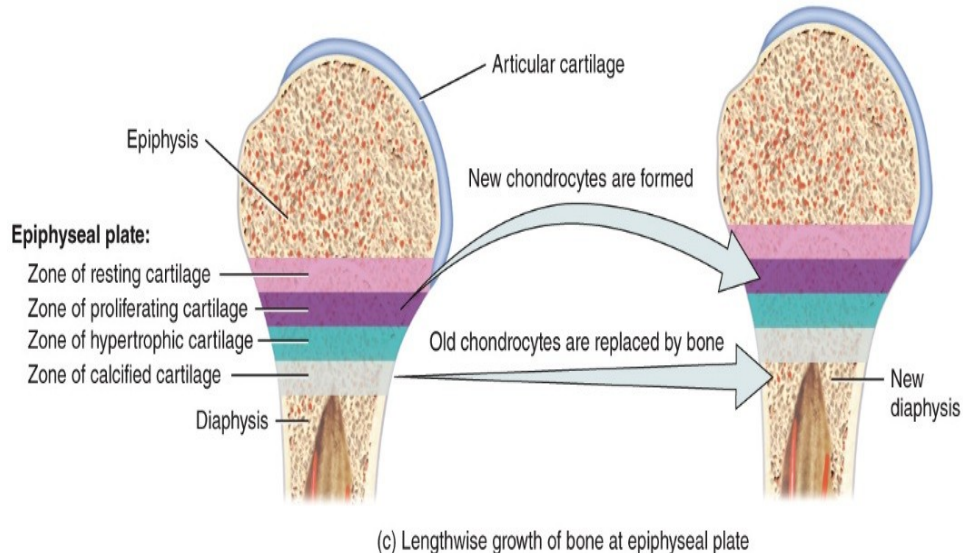


Scott Camazine/Science Source

(b) Twelve-week-old fetus. The red areas represent bones that are forming (calcified). Clear areas represent cartilage (uncalcified).

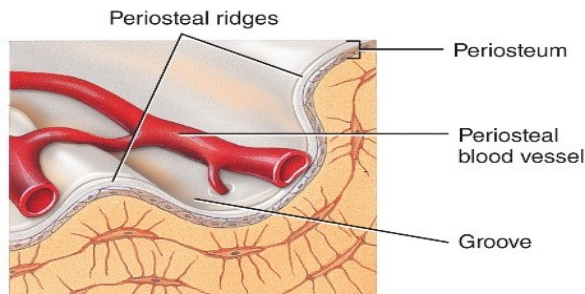
Bone Formation

- Endochondral ossification also occurs in epiphyseal plates of long bones as they grow in length

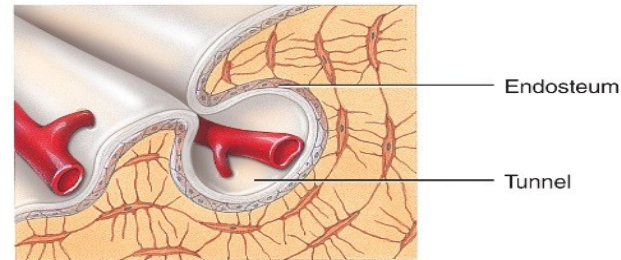


Bone Formation

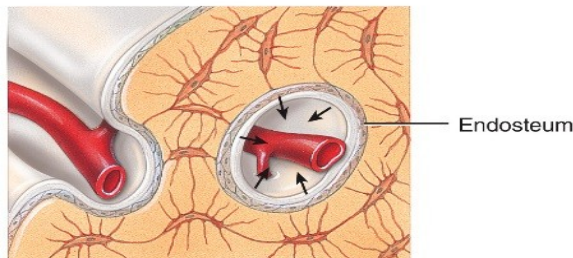
- Bones thicken thanks to the cooperative action of osteoblasts and osteoclasts



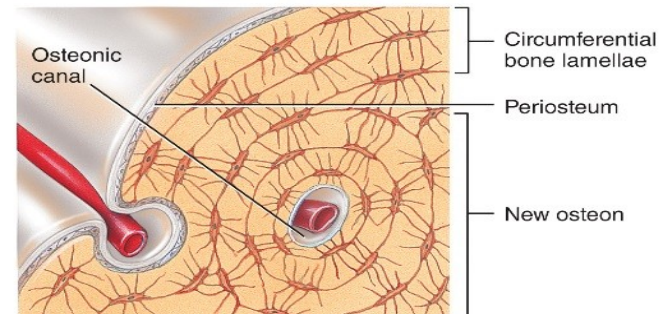
1 Ridges in periosteum create groove for periosteal blood vessel.



2 Periosteal ridges fuse, forming an endosteum-lined tunnel.



3 Osteoblasts in endosteum build new concentric bone lamellae inward toward center of tunnel, forming a new osteon.

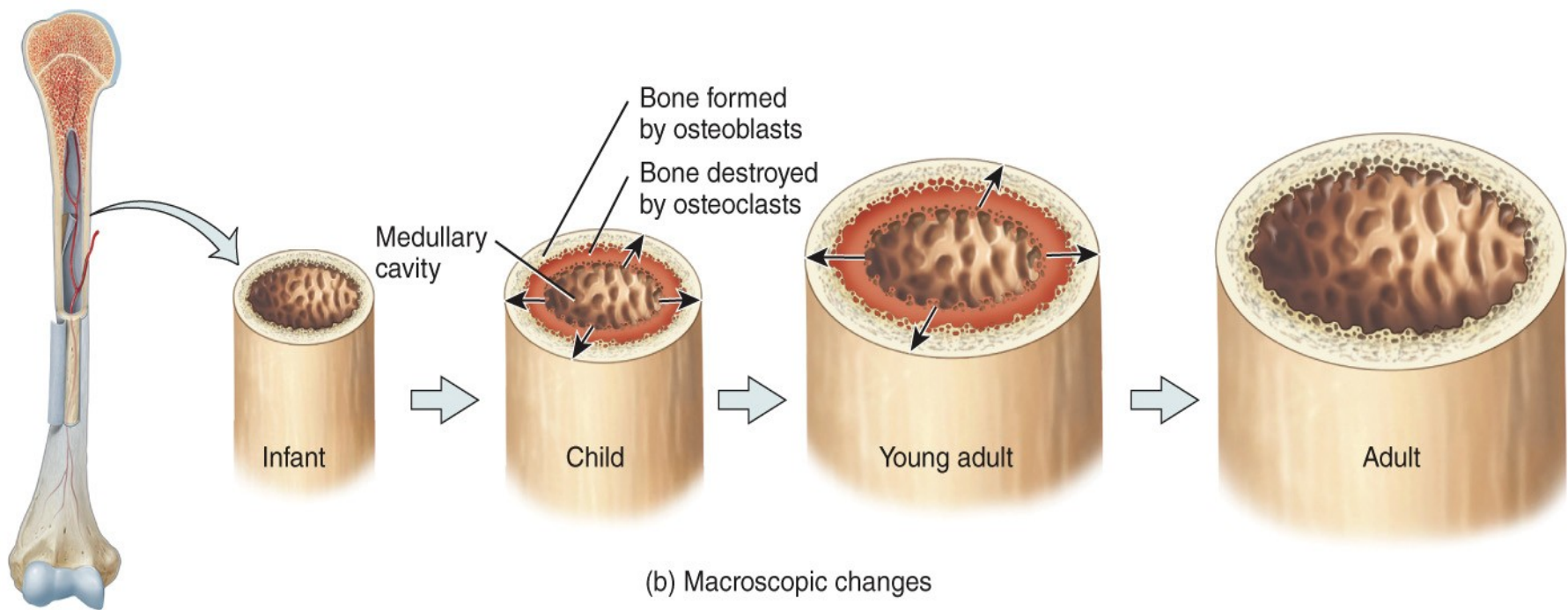


4 Bone grows outward as osteoblasts in periosteum build new circumferential bone lamellae. Osteon formation repeats as new periosteal ridges fold over blood vessels.

(a) Microscopic details

Bone Formation

- As osteoblasts deposit bone on the outer surface osteoclasts widen the medullary cavity from within



Bone Growth and Remodeling Interactions Animation:

Bone Growth and Remodeling – necessary nutrients

Bone Growth and Remodeling – maintenance of bone thickness

Factors Affecting Bone Growth and Remodeling

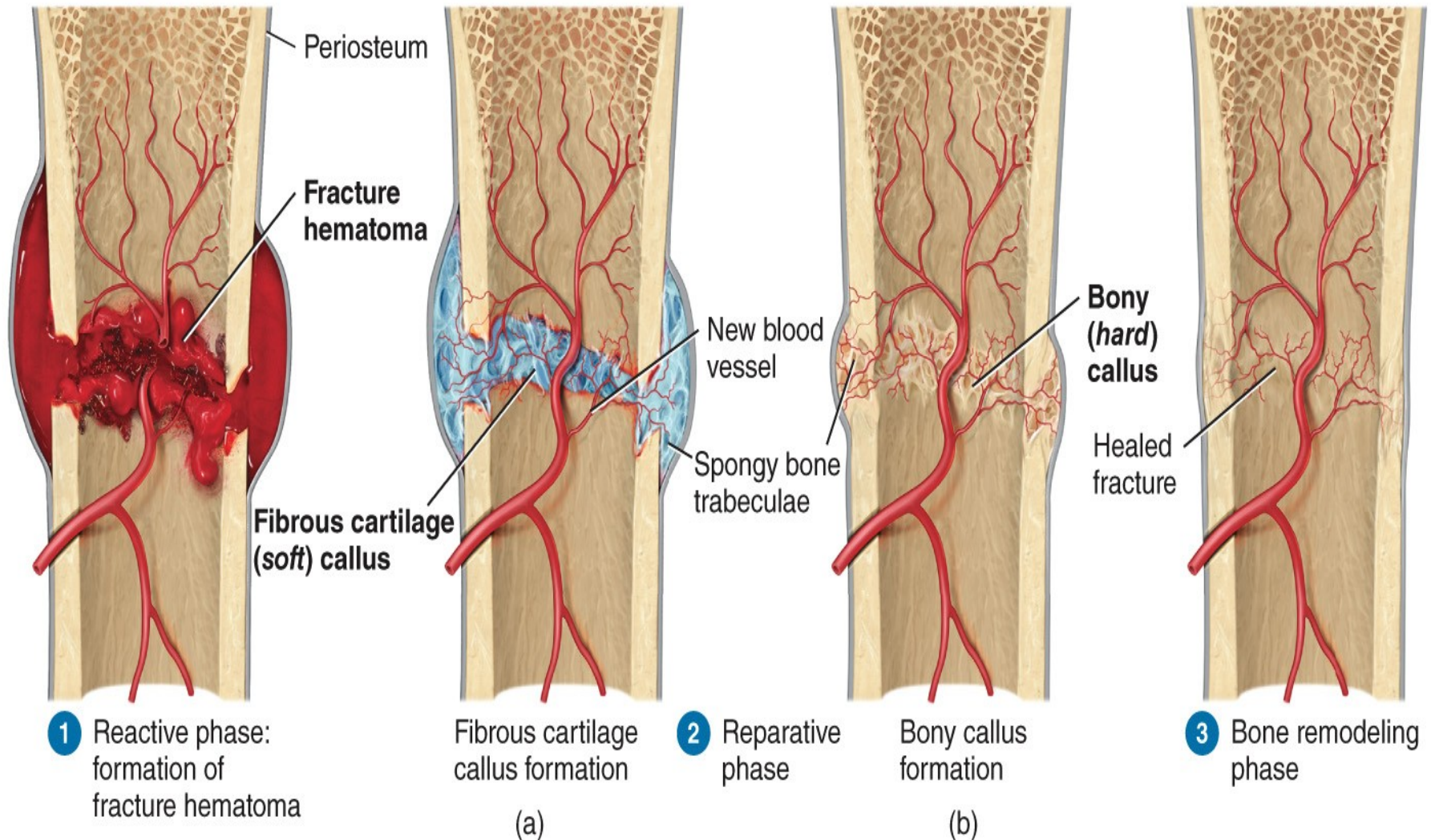
- Minerals – calcium, phosphorus, magnesium, fluoride, manganese
- Vitamins – Vitamins A, C, D, K and B₁₂
- Hormones – IGFs (stimulated by GH), T₃ and T₄, sex hormones (estrogen in females, testosterone in males)

Fracture and Repair of Bone

Fracture and Repair of Bone

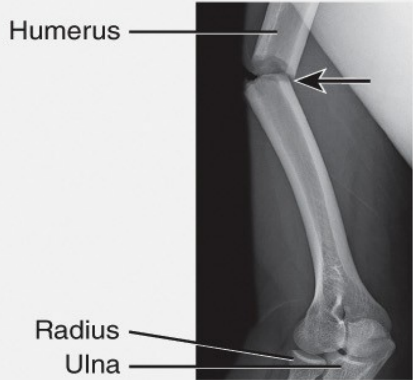
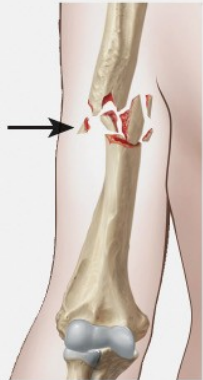
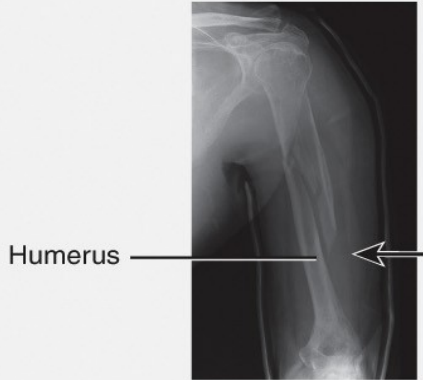
- A fracture is a break in a bone. Many different types of fractures may occur. The healing process involves 3 different phases:
 - The reactive phase is an early inflammatory phase
 - The reparative phase includes formation of a fibrocartilaginous callus first and a bony callus second
 - The bone remodeling phase is the last step as the bony callus is remodeled

Fracture and Repair of Bone



Fracture and Repair of Bone

TABLE 6.1 Some Common Fractures

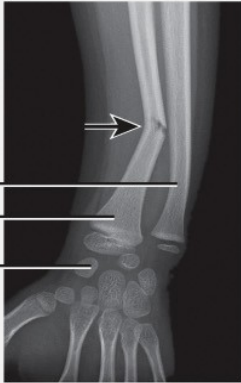
Fracture	Description	Illustration	Radiograph
Open (<i>Compound</i>)	The broken ends of the bone protrude through the skin. Conversely, a <i>closed (simple)</i> fracture does not break the skin.		 Humerus Radius Ulna
Comminuted (KOM-i-noo-ted; <i>com-</i> = together; <i>-minuted</i> = crumbled)	The bone is splintered, crushed, or broken into pieces at the site of impact, and smaller bone fragments lie between the two main fragments.		 Humerus

Courtesy Dr. Brent Layton

Courtesy Per Amundson, M.D.

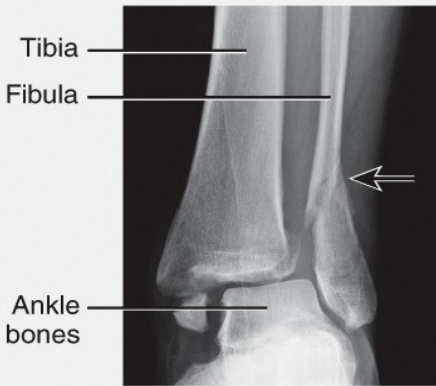
Fracture and Repair of Bone

TABLE 6.1 Some Common Fractures

Fracture	Description	Illustration	Radiograph
Greenstick	A partial fracture in which one side of the bone is broken and the other side bends, similar to the way a green twig breaks on one side while the other side stays whole, but bends; occurs only in children, whose bones are not fully ossified and contain more organic material than inorganic material.	 Ulna Radius Wrist bones	 Courtesy Dr. Brent Layton
Impacted	One end of the fractured bone is forcefully driven into the interior of the other.	 Humerus	 Courtesy Dr. Brent Layton

Fracture and Repair of Bone

TABLE 6.1 Some Common Fractures

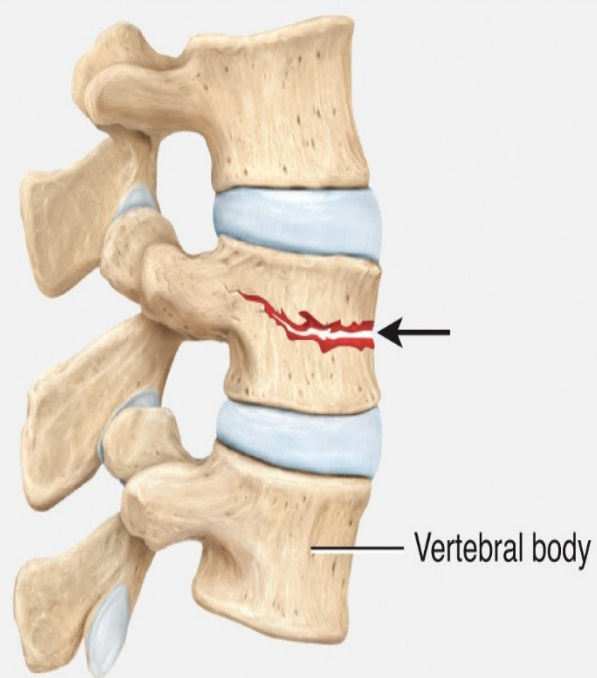
Fracture	Description	Illustration	radiograph
Pott	Fracture of the distal end of the fibula (lateral leg bone), with serious injury of the distal tibial articulation.		 Tibia Fibula Ankle bones
Colles (KOL-ēz)	Fracture of the distal end of the lateral forearm bone (radius) in which the distal fragment is displaced posteriorly.		 Radius Ulna Wrist bones

Courtesy Dr. Brent Layton

David J. Sartoris/bonepit.com

Fracture and Repair of Bone

TABLE 6.1 Some Common Fractures

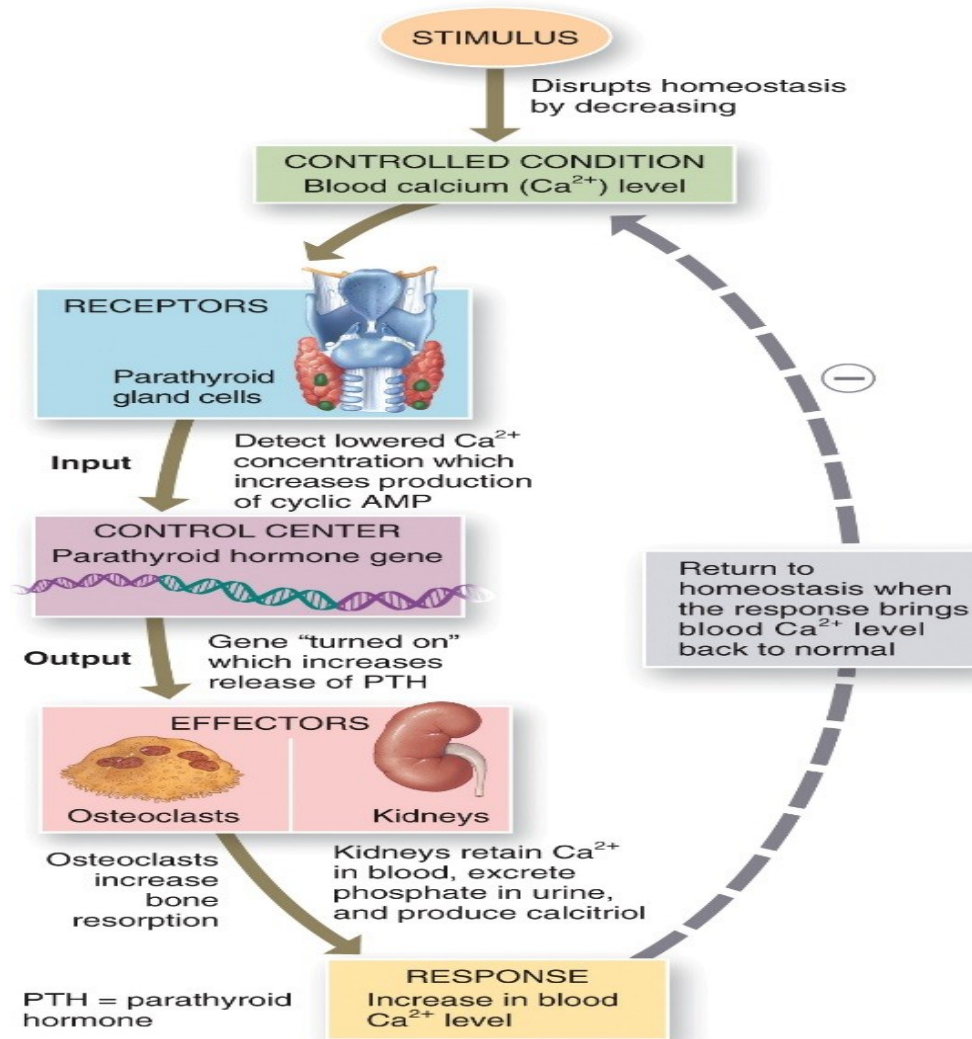
Fracture	Description	Illustration	radiograph
Vertebral compression fracture (VCF)	The vertebral body of one or more vertebrae fractures and becomes compressed into a wedge-shape. May be caused by injury, trauma, or more commonly in individuals with osteoporosis.		

Bone's Role in Calcium Homeostasis

Bone's Role in Calcium Homeostasis

- Bones store 99% of the body's calcium
- Parathyroid gland secretes parathyroid hormone (PTH) when calcium levels drop
 - Osteoclasts are stimulated to increase bone resorption and calcium is released
 - PTH also stimulates the production of calcitriol by the kidneys to increase calcium absorption in the intestines

Bone's Role in Calcium Homeostasis



Calcium Homeostasis Interactions Animation:

Calcium Homeostasis

Exercise and Aging in Bone Tissue

Exercise and Bone Tissue Interactions Animation:

Effect of Exercise on Bone

Aging and Bone Tissue

- From birth through adolescence, more bone is produced than is lost during remodeling
- In adults, the rates are the same
- Older individuals, especially post-menopausal women, experience a decrease in bone mass when resorption outpaces deposition



SEM 30x

(a) Normal bone



SEM 30x

(b) Osteoporotic bone

Summary of Factors That Affect Bone Growth: Minerals

Factor	Comment
Calcium and phosphorus	Make bone extracellular matrix hard.
Magnesium	Helps form bone extracellular matrix.
Fluoride	Helps strengthen bone extracellular matrix.
Manganese	Activates enzymes involved in synthesis of bone extracellular matrix.

Summary of Factors That Affect Bone Growth: Vitamins

Factor	Comment
Vitamin A	Needed for the activity of osteoblasts during remodeling of bone; deficiency stunts bone growth; toxic in high doses.
Vitamin C	Needed for synthesis of collagen, the main bone protein; deficiency leads to decreased collagen production, which slows down bone growth and delays repair of broken bones.
Vitamin D	Active form (calcitriol) is produced by the kidneys; helps build bone by increasing absorption of calcium from digestive canal into blood; deficiency causes faulty calcification and slows down bone growth; may reduce the risk of osteoporosis but is toxic if taken in high doses. People who have minimal exposure to ultraviolet rays or do not take vitamin D supplements may not have sufficient vitamin D to absorb calcium. This interferes with calcium metabolism.
Vitamins K and B ₁₂	Needed for synthesis of bone proteins; deficiency leads to abnormal protein production in bone extracellular matrix and decreased bone density.

Summary of Factors That Affect Bone Growth: Hormones

Factor	Comment
Growth hormone (GH)	Secreted by the anterior lobe of the pituitary gland; promotes general growth of all body tissues, including bone, mainly by stimulating production of insulin-like growth factors.
Insulin-like growth factors (IGFs)	Secreted by the liver, bones, and other tissues on stimulation by growth hormone; promotes normal bone growth by stimulating osteoblasts and by increasing the synthesis of proteins needed to build new bone.
Thyroid hormones (T_3 and T_4)	Secreted by thyroid gland; promote normal bone growth by stimulating osteoblasts.
Insulin	Secreted by the pancreas; promotes normal bone growth by increasing the synthesis of bone proteins.

Summary of Factors That Affect Bone Growth: Hormones

Factor	Comment
Sex hormones (estrogens and testosterone)	Secreted by the ovaries in women (estrogens) and by the testes in men (testosterone); stimulate osteoblasts and promote the sudden “growth spurt” that occurs during the teenage years; shut down growth at the epiphyseal plates around age 18–21, causing lengthwise growth of bone to end; contribute to bone remodeling during adulthood by slowing bone resorption by osteoclasts and promoting bone deposition by osteoblasts.
Parathyroid hormone (PTH)	Secreted by the parathyroid glands; promotes bone resorption by osteoclasts; enhances recovery of calcium ions from urine; promotes formation of the active form of vitamin D (calcitriol).
Calcitonin (CT)	Secreted by the thyroid gland; inhibits bone resorption by osteoclasts.

Summary of Factors That Affect Bone Growth: Exercise and Aging

Factor	Comment
Exercise	Weight-bearing activities stimulate osteoblasts and, consequently, help build thicker, stronger bones and retard loss of bone mass that occurs as people age.
Aging	As the level of sex hormones diminishes during middle age to older adulthood, especially in women after menopause, bone resorption by osteoclasts outpaces bone deposition by osteoblasts, which leads to a decrease in bone mass and an increased risk of osteoporosis.

Disorders: Homeostatic Imbalances

- Osteoporosis – bone resorption outpaces formation, 80% of those affected are women
- Rickets and Osteomalacia – inadequate calcification of extracellular bone matrix.
 - Rickets affects children and leads to bowed legs and deformations in skull, rib cage or pelvis
 - Osteomalacia affects adults and causes painful/tender bones and fractures with minor trauma

Medical Terminology

- Osteoarthritis – degeneration of articular cartilage, leads to friction of bone against bone
- Osteomyelitis – infection of bone often caused by *Staphylococcus aureus*
- Osteopenia – reduced bone mass below normal
- Osteosarcoma – bone cancer that primarily affects osteoblasts

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